

DSA Project

Ride Sharing System

Black Box Testing

1. Equivalence Partitioning

1. Driver Registration - Name Validation

Function Used:

isName(name)

Condition:

- Only alphabets allowed

Test Case ID	Input	Equivalence Class	Expected Result	Actual Result
EP-TC01	Ali	Valid alphabetic name	Accept	Accept
EP-TC02	AhmedKhan	Valid alphabetic name	Accept	Accept
EP-TC03	Ali123	Invalid (contains numbers)	Reject	Reject
EP-TC04	@Ali	Invalid (special character)	Reject	Reject
EP-TC05	12345	Invalid numeric input	Reject	Reject

```

Welcome to Elite Ride!
Thank you for choosing us

MAIN MENU

1. Rider
2. Driver
3. Admin

Choose an option: 2

Register yourself as a driver!

Enter your name: Ali123
invalid name! name must only be alphabetic
Enter Name: @Ali
invalid name! name must only be alphabetic
Enter Name: 12345
invalid name! name must only be alphabetic
Enter Name: AhmedKhan
Enter Phone Number: _
```

2. Phone Number Validation

Condition:

- Length exactly 11 digits

Test Case ID	Input	Equivalence Class	Expected Result	Actual Result
EP-TC06	03123456789	Valid 11-digit number	Accept	Accept
EP-TC07	0300123456	Invalid (<11 digits)	Reject	Reject
EP-TC08	031234567890	Invalid (>11 digits)	Reject	Reject
EP-TC09	abcdefghijkl	Invalid alphabetic input	Reject	Accept
EP-TC10	03@12345678	Invalid special character	Reject	Accept

```

Welcome to Elite Ride!
Thank you for choosing us

MAIN MENU

1. Rider
2. Driver
3. Admin

Choose an option: 2
Register yourself as a driver!
Enter your name: faheem
Enter Phone Number: 0300123456
Wrong number! Length must be 11 digits
Enter Phone Numbe: 031234567890
Wrong number! Length must be 11 digits
Enter Phone Numbe: 03@12345678
Enter Email: _
```

```

Welcome to Elite Ride!
Thank you for choosing us

MAIN MENU

1. Rider
2. Driver
3. Admin

Choose an option: 2
Register yourself as a driver!
Enter your name: faheem
Enter Phone Number: abcdefghijk
Enter Email:
```

3. Email Validation

Function:

is_valid(email)

Conditions:

- Must contain @
- Must contain .

- First character alphabetic

Test Case ID	Input	Equivalence Class	Expected Result	Actual Result
EP-TC11	ali@gmail.com	Valid email	Accept	Accept
EP-TC12	a123@yahoo.com	Valid email	Accept	Accept
EP-TC13	aligmail.com	Missing @	Reject	Reject
EP-TC14	ali@gmailcom	Missing dot	Reject	Reject
EP-TC15	1ali@gmail.com	First char not alphabetic	Reject	Reject
EP-TC16	ali@.com	Invalid format	Reject	Accept

```

Welcome to Elite Ride!
Thank you for choosing us

MAIN MENU

1. Rider
2. Driver
3. Admin

Choose an option: 2

Register yourself as a driver!

Enter your name: faheem
Enter Phone Number: 03123456789
Enter Email: aligmail.com
Wrong email provided!
Enter correct Email: ali@gmailcom
Wrong email provided!
Enter correct Email: 1ali@gmail.com
Wrong email provided!
Enter correct Email: ali@.com
Enter Age: 21
```

```

Welcome to Elite Ride!
Thank you for choosing us

MAIN MENU

1. Rider
2. Driver
3. Admin

Choose an option: 2

Register yourself as a driver!

Enter your name: shawa
Enter Phone Number: 03123456789
Enter Email: ali@gmail.com
Enter Age: 21
```

4. Age Validation

Condition:

- Age must be ≥ 18

Test Case ID	Input	Equivalence Class	Expected Result	Actual Result
EP-TC17	18	Valid age	Register	Register

EP-TC18	25	Valid age	Register	Register
EP-TC19	17	Invalid (<18)	Reject	Reject
EP-TC20	0	Invalid unrealistic age	Reject	Reject
EP-TC21	-5	Invalid negative age	Reject	Reject

Welcome to Elite Ride!
Thank you for choosing us

MAIN MENU

1. Rider
2. Driver
3. Admin

Choose an option: 2

Register yourself as a driver!

Enter your name: faheem
Enter Phone Number: 03123456789
Enter Email: faah@gmail.com
Enter Age: 17
Not old enough to register

1. Main menu
0. Exit

Choose an option: _

Welcome to Elite Ride!
Thank you for choosing us

MAIN MENU

1. Rider
2. Driver
3. Admin

Choose an option: 2

Register yourself as a driver!

Enter your name: fah
Enter Phone Number: 03123456789
Enter Email: ali@gmail.com
Enter Age: -5
Not old enough to register

1. Main menu
0. Exit

Choose an option:

```
MAIN MENU

1. Rider
2. Driver
3. Admin

Choose an option: 2

Register yourself as a driver!

Enter your name: faheem
Enter Phone Number: 03123456789
Enter Email: ali@gmail.com
Enter Age: 24
Enter NIC: _
```

5. NIC Validation

Condition:

- NIC length must be exactly 13 digits

Test Case ID	Input	Equivalence Class	Expected Result	Actual Result
EP-TC22	3520212345671	Valid NIC	Accept	Accept
EP-TC23	35202123456	Invalid (<13 digits)	Reject	Reject
EP-TC24	352021234567123	Invalid (>13 digits)	Reject	Reject
EP-TC25	abcdef1234567	Invalid alphabets	Reject	Accept


```

1. Rider
2. Driver
3. Admin

Choose an option: 2

Register yourself as a driver!

Enter your name: faheem
Enter Phone Number: 03123456789
Enter Email: ali@gmail.com
Enter Age: 24
Enter NIC: 3520212345671

You have been registered!

1. Main menu
0. Exit

Choose an option:

```

```

MAIN MENU

1. Rider
2. Driver
3. Admin

Choose an option: 2

Register yourself as a driver!

Enter your name: faheem
Enter Phone Number: 03123456789
Enter Email: ali@gmail.com
Enter Age: 24
Enter NIC: 35202123456
Wrong NIC! Length must be 13 digits
Enter NIC: 352021234567123
Wrong NIC! Length must be 13 digits
Enter NIC: abcdef1234567

You have been registered!

1. Main menu
0. Exit

Choose an option:

```

6. Admin Login Validation

Condition:

- Username = dsaa
- Password = dsaa.project

Test Case ID	Input	Equivalence Class	Expected Result	Actual Result
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EP-TC26	Correct username & password	Valid credentials	Login success	Login success
EP-TC27	Wrong username	Invalid username	Reject	Accept proceed to password
EP-TC28	Wrong password	Invalid password	Reject	Reject
EP-TC29	Both wrong	Invalid credentials	Reject	Reject
EP-TC30	Empty username/password	Invalid empty input	Reject	Do nothing

```
LOG IN
Enter username: dsaa
Enter password: dsa.project
Wrong info, please try again!

LOG IN
Enter username: dsaa
Enter password: dsaa.project
You're logged in!

ADMIN MENU

1. View Driver Records
2. View Riders Records
3. View Feedback
4. Main Menu / Exit

Choose an option: _
```

```
Welcome to Elite Ride!
Thank you for choosing us

MAIN MENU

1. Rider
2. Driver
3. Admin

Choose an option: 3

LOG IN
Enter username: faheem
Enter password: 1122
Wrong info, please try again!

LOG IN
Enter username:
dsa
Enter password:
dsa
Wrong info, please try again!

LOG IN
Enter username: dsaa
Enter password: hello
Wrong info, please try again!
```

7. Pickup & Destination Location Validation

Condition:

- Valid locations from 1–13

Test Case ID	Input	Equivalence Class	Expected Result	Actual Result
EP-TC31	1	Valid location	Accept	Accept
EP-TC32	13	Valid location	Accept	Accept
EP-TC33	0	Invalid (<1)	Reject	Reject
EP-TC34	14	Invalid (>13)	Reject	Reject
EP-TC35	-2	Invalid negative value	Reject	Reject

```

lets book a ride for you!
Enter the number of passengers: 4
A Silver coloured Hyundai will be waiting right at your location!
1)University Road
2)Karsaz
3)Bahadurabad
4)Gulshan
5)North Nazimabad
6)Baloch Colony
7)Defense
8)Clifton
9)Metropole
10)Kemari
11)Saddar
12)Gulberg
13)Lyari Expressway
Enter a number for Pickup location(1-13): 0
Incorrect location
Enter a number for Pickup location(1-13): 15
Incorrect location
Enter a number for Pickup location(1-13): -3
Incorrect location
Enter a number for Pickup location(1-13): 5
Enter a number for Destination(1-13): 0
Incorrect location
Enter a number for Destination(1-13): 14
Incorrect location
Enter a number for Destination(1-13): -9
Incorrect location
Enter a number for Destination(1-13): 2
Your car is booked!
Your Drivers name: Aishafaheem
Ride ID: 13070
The total fare from North Nazimabad to Karsaz is 700Rs

```

8. Feedback Rating Validation

Condition:

- Rating between 1–5

Test Case ID	Input	Equivalence Class	Expected Result
EP-TC36	5	Valid rating	Accept

EP-TC37	3	Valid rating	Accept
EP-TC38	0	Invalid (<1)	Reject
EP-TC39	6	Invalid (>5)	Reject
EP-TC40	-1	Invalid negative value	Reject

Note:

The problem is that feedback function is never called anywhere in the menus.
So users cannot actually give feedback.

9. Car Passenger Capacity Validation

Condition:

- Car assigned according to passenger count

Test Case ID	Input	Equivalence Class	Expected Result	Actual Result
EP-TC41	1	Valid passenger count	Bike assigned	Bike assigned
EP-TC42	4	Valid passenger count	Suitable car assigned	Suitable car assigned
EP-TC43	15	Maximum valid passengers	Large car assigned	Large car assigned
EP-TC44	16	Invalid (>capacity)	No car available	No car available
EP-TC45	0	Invalid zero passengers	Reject	Bike assigned

```
Lets book a ride for you!
```

```
Enter the number of passengers: 1
```

```
A Red coloured Heavy Bike will be waiting right at your location!
```

```
Lets book a ride for you!
```

```
Enter the number of passengers: 4
```

```
A Silver coloured Hyundai will be waiting right at your location!
```

```
Lets book a ride for you!

Enter the number of passengers: 15

A Red coloured Honda City will be waiting right at your location!

Choose an option: 1

Lets book a ride for you!

Enter the number of passengers: 16
No car available for given number of passengers!

Press any key to continue
_

Choose an option: 1

Lets book a ride for you!

Enter the number of passengers: 0

A Red coloured Heavy Bike will be waiting right at your location!
```

2. Boundary Value Analysis

1. Age Validation (Driver & Rider Registration)

Condition: Age must be ≥ 18

Test Case	Input	Boundary Type	Expected Result	Expected Result
BVA01	17	Just below lower boundary	Reject	Reject
BVA02	18	Lower boundary (valid)	Accept	Accept
BVA03	19	Just above lower boundary	Accept	Accept
BVA04	0	Extreme invalid lower value	Reject	Reject
BVA05	100	Valid upper extreme (no upper limit assumed)	Accept	Accept

Welcome to Elite Ride!
Thank you for choosing us

MAIN MENU

1. Rider
2. Driver
3. Admin

Choose an option: 2

Register yourself as a driver!

Enter your name: fah
Enter Phone Number: 03001234567
Enter Email: fahe@gmail.com
Enter Age: 17
Not old enough to register

1. Main menu
0. Exit

Choose an option: _

Welcome to Elite Ride!
Thank you for choosing us

MAIN MENU

1. Rider
2. Driver
3. Admin

Choose an option: 2

Register yourself as a driver!

Enter your name: fah
Enter Phone Number: 03123456789
Enter Email: ali@gmail.com
Enter Age: -5
Not old enough to register

1. Main menu
0. Exit

Choose an option:

```

Welcome to Elite Ride!
Thank you for choosing us

MAIN MENU

1. Rider
2. Driver
3. Admin

Choose an option: 2

Register yourself as a driver!

Enter your name: faheem
Enter Phone Number: 03001234567
Enter Email: fah@gmail.com
Enter Age: 100
Enter NIC:
```

2. Phone Number Validation (Length = 11 digits)

Test Case	Input	Boundary Type	Expected Result	Actual Result
BVA06	10 digits	Just below lower boundary	Reject	Reject
BVA07	11 digits	Exact boundary (valid)	Accept	Accept
BVA08	12 digits	Just above upper boundary	Reject	Reject
BVA09	alphabets input	Invalid non-numeric input	Reject	Accept
BVA10	special characters	Invalid format input	Reject	Accept

Welcome to Elite Ride!
Thank you for choosing us

MAIN MENU

1. Rider
2. Driver
3. Admin

Choose an option: 2

Register yourself as a driver!

Enter your name: fah

Enter Phone Number: 03001234567

Enter Email: _

Welcome to Elite Ride!
Thank you for choosing us

MAIN MENU

1. Rider
2. Driver
3. Admin

Choose an option: 2

Register yourself as a driver!

Enter your name: faheem

Enter Phone Number: 0300123456

Wrong number! Length must be 11 digits

Enter Phone Numbe: 030012345678

Wrong number! Length must be 11 digits

Enter Phone Numbe: 030012345a

Wrong number! Length must be 11 digits

Enter Phone Numbe: 030012345aa

Enter Email:

```

Welcome to Elite Ride!
Thank you for choosing us

MAIN MENU

1. Rider
2. Driver
3. Admin

Choose an option: 2

Register yourself as a driver!

Enter your name: faheem
Enter Phone Number: 0300123456*
Enter Email: 
```

3. NIC Validation (Length = 13 digits)

Test Case	Input	Boundary Type	Expected Result
BVA11	12 digits	Just below boundary	Reject
BVA12	13 digits	Exact boundary (valid)	Accept
BVA13	14 digits	Just above boundary	Reject

Welcome to Elite Ride!
Thank you for choosing us

MAIN MENU

1. Rider
2. Driver
3. Admin

Choose an option: 2

Register yourself as a driver!

Enter your name: faheem

Enter Phone Number: 03001234567

Enter Email: fah@gmail.com

Enter Age: 100

Enter NIC: 352021234567

Wrong NIC! Length must be 13 digits

Enter NIC:

 C:\Users\user\Downloads\Elite-Drive-main\Elite-Drive-main\EliteDrive.exe

Welcome to Elite Ride!
Thank you for choosing us

MAIN MENU

1. Rider
2. Driver
3. Admin

Choose an option: 2

Register yourself as a driver!

Enter your name: faheem

Enter Phone Number: 03001234567

Enter Email: fah@gmail.com

Enter Age: 100

Enter NIC: 352021234567

Wrong NIC! Length must be 13 digits

Enter NIC: 35202123456711

Wrong NIC! Length must be 13 digits

Enter NIC:

```
Register yourself as a driver!

Enter your name: faheem
Enter Phone Number: 03001234567
Enter Email: fah@gmail.com
Enter Age: 100
Enter NIC: 352021234567
Wrong NIC! Length must be 13 digits
Enter NIC: 35202123456711
Wrong NIC! Length must be 13 digits
Enter NIC: 3520212345671

You have been registered!

1. Main menu
0. Exit
```

4. Rating System (Range 1 to 5)

Test Case	Input	Boundary Type	Expected Result
BVA14	0	Just below lower boundary	Reject
BVA15	1	Lower boundary (valid)	Accept
BVA16	2	Inside valid range	Accept
BVA17	4	Inside valid range	Accept
BVA18	5	Upper boundary (valid)	Accept
BVA19	6	Just above upper boundary	Reject

Note:

The problem is that feedback function is never called anywhere in the menus.
So users cannot actually give feedback.

5. Location Selection (Range 1 to 13)

Test Case	Input	Boundary Type	Expected Result	Actual Result
BVA20	0	Just below lower boundary	Reject	Reject
BVA21	1	Lower boundary (valid)	Accept	Accept

BVA22	13	Upper boundary (valid)	Accept	Accept
BVA23	14	Just above upper boundary	Reject	Reject

Enter the number of passengers: 5

A White coloured Audi will be waiting right at your location!

- 1)University Road
- 2)Karsaz
- 3)Bahadurabad
- 4)Gulshan
- 5)North Nazimabad
- 6)Baloch Colony
- 7)Defense
- 8)Clifton
- 9)Metropole
- 10)Kemari
- 11)Saddar
- 12)Gulberg
- 13)Lyari Expressway

Enter a number for Pickup location(1-13): 0

Incorrect location

Enter a number for Pickup location(1-13):

Enter the number of passengers: 5

A White coloured Audi will be waiting right at your location!

- 1)University Road
- 2)Karsaz
- 3)Bahadurabad
- 4)Gulshan
- 5)North Nazimabad
- 6)Baloch Colony
- 7)Defense
- 8)Clifton
- 9)Metropole
- 10)Kemari
- 11)Saddar
- 12)Gulberg
- 13)Lyari Expressway

Enter a number for Pickup location(1-13): 0

Incorrect location

Enter a number for Pickup location(1-13): 15

Incorrect location

Enter a number for Pickup location(1-13):

```
3)Bahadurabad
4)Gulshan
5)North Nazimabad
6)Baloch Colony
7)Defense
8)Clifton
9)Metropole
10)Kemari
11)Saddar
12)Gulberg
13)Lyari Expressway

Enter a number for Pickup location(1-13): 0
Incorrect location
Enter a number for Pickup location(1-13): 15
Incorrect location
Enter a number for Pickup location(1-13): 4
Enter a number for Destination(1-13): 0
Incorrect location
Enter a number for Destination(1-13): 1

    Your car is booked!

Your Drivers name: faheem
Ride ID: 29147
The total fare from Gulshan to University road is 300Rs
```

6. Passenger Count (Car Assignment)

Test Case	Input	Boundary Type	Expected Result	Actual Result
BVA24	0	Invalid lower boundary	Reject	Bike assigned
BVA25	1	Minimum valid boundary	Bike assigned	Bike assigned
BVA26	4	Small car range boundary	Small car assigned	Small car assigned
BVA27	10	Medium range boundary	Medium car assigned	Medium car assigned
BVA28	15	Upper valid boundary	Large vehicle assigned	Large vehicle assigned
BVA29	16	Just above upper boundary	No car available	No car available

```
You are Registered!

Lets book a ride for you!

Enter the number of passengers: 16
No car available for given number of passengers!
```

```

RIDER MENU

1. Search for another ride
2. See all accepted rides
3. Cancel a ride
4. Main Menu / Exit

Choose an option:
```

```
Lets book a ride for you!

Enter the number of passengers: 0

A Red coloured Heavy Bike will be waiting right at your location!
```

```
Lets book a ride for you!

Enter the number of passengers: 1

A Red coloured Heavy Bike will be waiting right at your location!
```

```
Lets book a ride for you!

Enter the number of passengers: 10

A White coloured Toyota Innova will be waiting right at your location!
```

7. Admin Login Validation

Test Case	Input	Boundary Type	Expected Result	Actual Result
BVA30	empty username/password	Null input boundary	Reject	Nothing happened
BVA31	wrong username	Invalid input case	Reject	Reject
BVA32	wrong password	Invalid input case	Reject	Reject
BVA33	correct credentials	Valid authentication boundary	Accept	Accept

Welcome to Elite Ride!
Thank you for choosing us

MAIN MENU

1. Rider
2. Driver
3. Admin

Choose an option: 3

LOG IN

Enter username:

dsaa

Enter password:

LOG IN

Enter username: ahmad

Enter password: 1123

Wrong info, please try again!

LOG IN

Enter username: _

```
LOG IN

Enter username: ahmad
Enter password: 1123
Wrong info, please try again!

LOG IN

Enter username: dsaa
Enter password: dsaa.project
You're logged in!

ADMIN MENU

1. View Driver Records
2. View Riders Records
3. View Feedback
4. Main Menu / Exit

Choose an option:
```

3. Decision Table Testing

1. Admin Login Decision Table

Conditions:

- C1: Username correct?
- C2: Password correct?

Actions:

- A1: Login success
- A2: Login fail

Conditions	R1	R2	R3	R4
Username correct	Yes	Yes	No	No
Password correct	Yes	No	Yes	No
Actions				
Login Success	Yes	No	No	No

Login Fail	No	Yes	Yes	Yes
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2. Driver Registration Decision Table

Conditions:

- C1: Name valid (alphabetic)
- C2: Phone valid (11 digits)
- C3: Email valid
- C4: Age ≥ 18
- C5: NIC valid (13 digits)

Actions:

- A1: Register driver
- A2: Reject registration

Conditions	R1	R2	R3	R4	R5
Name valid	Y	Y	Y	Y	N
Phone valid	Y	Y	N	Y	Y
Email valid	Y	N	Y	Y	Y
Age ≥ 18	Y	Y	Y	N	Y
NIC valid	Y	Y	Y	Y	Y
Actions					
Register	Yes	No	No	No	No
Reject	No	Yes	Yes	Yes	Yes

3. Rider Registration Decision Table

Conditions:

- C1: Name valid
- C2: Phone valid
- C3: Email valid

- C4: Age ≥ 18
- C5: NIC valid

Actions:

- A1: Register rider
- A2: Reject

Conditions	R1	R2	R3	R4	R5	R6
Name Valid	Yes	No	Yes	Yes	Yes	Yes
Phone Valid	Yes	Yes	No	Yes	Yes	Yes
Email Valid	Yes	Yes	Yes	No	Yes	Yes
Age ≥ 18	Yes	Yes	Yes	Yes	No	Yes
NIC Valid	Yes	Yes	Yes	Yes	Yes	No
Actions						
Register Rider	Yes	No	No	No	No	No
Reject Registration	No	Yes	Yes	Yes	Yes	Yes

4. Ride Booking Decision Table

Conditions:

- C1: Valid pickup (1–13)
- C2: Valid destination (1–13)
- C3: Passenger count valid (>0)
- C4: Car available

Actions:

- A1: Book ride
- A2: Reject ride
- A3: Assign driver
- A4: Assign car

Conditions	R1	R2	R3	R4	R5
Pickup valid	Y	Y	Y	N	Y

Destination valid	Y	Y	N	Y	Y
Passenger valid	Y	N	Y	Y	Y
Car available	Y	Y	Y	Y	N
Actions					
Book ride	Yes	No	No	No	No
Assign driver	Yes	No	No	No	No
Assign car	Yes	No	No	No	No

5. Feedback System Decision Table

Conditions:

- C1: Rating between 1–5
- C2: Review provided

Actions:

- A1: Accept feedback
- A2: Reject feedback

Conditions	R1	R2	R3
Rating valid	Y	N	Y
Review present	Y	Y	N
Actions			
Accept feedback	Yes	No	No
Reject feedback	No	Yes	Yes

4. State Transition Testing

1. Rider Registration and Ride Booking State Transition

States

State ID	State Name
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S1	Main Menu
S2	Rider Registration
S3	Rider Registered
S4	Ride Booking
S5	Ride Confirmed
S6	Ride Cancelled
S7	Rider Menu
S8	Exit

State Transition Table

Current State	Events / Inputs	Next State	Expected Output
S1 Main Menu	Select Rider Option	S2 Rider Registration	Registration form displayed
S2 Rider Registration	Valid details entered	S3 Rider Registered	Registration successful
S2 Rider Registration	Invalid details entered	S2 Rider Registration	Error message shown
S3 Rider Registered	Book ride selected	S4 Ride Booking	Ride booking starts
S4 Ride Booking	Valid pickup/destination	S5 Ride Confirmed	Ride booked successfully
S4 Ride Booking	Invalid location entered	S4 Ride Booking	Error message displayed
S5 Ride Confirmed	Cancel ride selected	S6 Ride Cancelled	Ride removed
S5 Ride Confirmed	View rides selected	S7 Rider Menu	Ride list displayed
S7 Rider Menu	Exit selected	S8 Exit	System terminated

2. Admin Login State Transition Testing

States

State ID	State Name
A1	Login Screen

A2	Login Failed
A3	Admin Logged In
A4	Admin Menu
A5	Exit

State Transition Table

Current State	Event	Next State	Expected Result
A1 Login Screen	Correct username/password	A3 Admin Logged In	Login success
A1 Login Screen	Wrong credentials	A2 Login Failed	Error shown
A2 Login Failed	Retry login	A1 Login Screen	Login screen reopened
A3 Admin Logged In	Open admin menu	A4 Admin Menu	Menu displayed
A4 Admin Menu	Exit selected	A5 Exit	Program exits

3. Ride Cancellation State Transition

States

State ID	State Name
R1	Ride Active
R2	Ride Cancel Request
R3	Ride Cancelled
R4	Ride ID Not Found

State Transition Table

Current State	Input/Event	Next State	Expected Result
R1 Ride Active	Valid ride ID entered	R3 Ride Cancelled	Ride deleted
R1 Ride Active	Invalid ride ID entered	R4 Ride ID Not Found	Error shown

R4 Ride ID Not Found	Retry cancellation	R2 Ride Cancel Request	Ask for ride ID again
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5. Use Case Testing

Use Case 1 - Rider Registration

Field	Description
Use Case ID	UC-01
Use Case Name	Rider Registration
Actor	Rider
Precondition	User is on Main Menu
Main Flow	1. User selects Rider option 2. System asks for registration details 3. User enters name, phone, email, age, and NIC 4. System validates data 5. Rider registered successfully
Alternate Flow	Invalid name, email, phone number, age, or NIC entered
Postcondition	Rider details stored in file

Test Cases

Test Case ID	Input	Expected Result
UC01-T1	Valid rider details	Rider registered successfully
UC01-T2	Name contains digits	Error message displayed
UC01-T3	Phone number less than 11 digits	Error message displayed
UC01-T4	Invalid email format	Error message displayed
UC01-T5	Age below 18	Registration rejected
UC01-T6	NIC length not 13 digits	Error message displayed


```

Welcome to Elite Ride!
Thank you for choosing us

MAIN MENU

1. Rider
2. Driver
3. Admin

Choose an option: 2
Register yourself as a driver!

Enter your name: shawal
Enter Phone Number: 03001212121
Enter Email: shawlu@gmail.com
Enter Age: 21
Enter NIC: 3202212345678

You have been registered!

1. Main menu
0. Exit

Choose an option: 1
```

Use Case 2 - Driver Registration

Field	Description
Use Case ID	UC-02
Use Case Name	Driver Registration
Actor	Driver
Precondition	User is on Main Menu
Main Flow	1. User selects Driver option 2. System asks for driver details 3. Driver enters valid information 4. System validates data 5. Driver registered successfully
Alternate Flow	Invalid details entered
Postcondition	Driver information stored in files

```

Welcome to Elite Ride!
Thank you for choosing us

MAIN MENU

1. Rider
2. Driver
3. Admin

Choose an option: 2
Register yourself as a driver!
Enter your name: shawal
Enter Phone Number: 03001212121
Enter Email: shawlu@gmail.com
Enter Age: 21
Enter NIC: 3202212345678
You have been registered!

1. Main menu
0. Exit

Choose an option: _
```

Test Cases

Test Case ID	Input	Expected Result
UC02-T1	Valid driver details	Driver registered
UC02-T2	Invalid name	Validation error
UC02-T3	Invalid email	Validation error
UC02-T4	Phone number not 11 digits	Validation error
UC02-T5	Age below 18	Registration rejected
UC02-T6	NIC length incorrect	Validation error

Use Case 3 - Book Ride

Field	Description
Use Case ID	UC-03
Use Case Name	Book Ride
Actor	Rider
Precondition	Rider successfully registered

Main Flow	1. Rider enters passenger count 2. System assigns suitable car 3. Rider selects pickup and destination 4. Driver assigned 5. Fare generated
Alternate Flow	Invalid passenger count or invalid location
Postcondition	Ride booked successfully

Test Cases

Test Case ID	Input	Expected Result
UC03-T1	Passenger count = 2	Car assigned
UC03-T2	Passenger count = 15	Large car assigned
UC03-T3	Passenger count > 15	"No car available"
UC03-T4	Pickup location = 5	Accepted
UC03-T5	Pickup location = 14	Error displayed
UC03-T6	Destination location = 0	Error displayed
UC03-T7	Valid pickup and destination	Ride confirmed with fare

```

You are Registered!

Lets book a ride for you!

Enter the number of passengers: 7

A Black coloured BMW will be waiting right at your location!

1)University Road
2)Karsaz
3)Bahadurabad
4)Gulshan
5)North Nazimabad
6)Baloch Colony
7)Defense
8)Clifton
9)Metropole
10)Kemari
11)Saddar
12)Gulberg
13)Lyari Expressway

Enter a number for Pickup location(1-13): 2
Enter a number for Destination(1-13): 6

    Your car is booked!

```

```

Your Drivers name: Aishafaheem
Ride ID: 1372
The total fare from Karsaz to Baloch Colony is 1100Rs

```

```

    RIDER MENU

    1. Search for another ride
    2. See all accepted rides
    3. Cancel a ride
    4. Main Menu / Exit

    Choose an option: _

```

Use Case 4 - Admin Login

Field	Description
Use Case ID	UC-05
Use Case Name	Admin Login
Actor	Admin
Precondition	Admin credentials available
Main Flow	1. Admin enters username and password 2. System validates credentials 3. Admin menu displayed
Alternate Flow	Wrong username or password
Postcondition	Admin accesses records

Test Cases

Test Case ID	Input	Expected Result
UC05-T1	Correct username and password	Login successful
UC05-T2	Incorrect username	Error message
UC05-T3	Incorrect password	Error message
UC05-T4	Both username and password wrong	Login rejected

```

Welcome to Elite Ride!
Thank you for choosing us

MAIN MENU

1. Rider
2. Driver
3. Admin

Choose an option: 3

LOG IN
Enter username: dsaa
Enter password: dsaa.project
You're logged in!

ADMIN MENU

1. View Driver Records
2. View Riders Records
3. View Feedback
4. Main Menu / Exit

Choose an option:
```

Use Case 6 - View Records and Feedback

Field	Description
Use Case ID	UC-06
Use Case Name	View Records

Actor	Admin
Precondition	Admin logged in
Main Flow	1. Admin selects record option 2. System opens corresponding file 3. Records displayed
Alternate Flow	File does not exist
Postcondition	Records shown on screen

Test Cases

Test Case ID	Input	Expected Result
UC06-T1	View Driver Records	Driver data displayed
UC06-T2	View Rider Records	Rider data displayed
UC06-T3	View Feedback	Feedback displayed
UC06-T4	Missing file	“File doesn’t exist” message

ADMIN MENU

1. View Driver Records
2. View Riders Records
3. View Feedback
4. Main Menu / Exit

Choose an option: 1

DRIVER RECORDS

Name: faheem
Phone Number: 03121234567
Email: asc@gmail.com
NIC: 1212121212121
Age: 20

Name: faheem
Phone Number: 03@12345678
Email: f@gmail.com
NIC: 3520212345671
Age: 32

Name: faheem
Phone Number: 03123456789
Email: ali@gmail.com
NIC: abcdef1234567
Age: 24

Name: faheem
Phone Number: 03123456789
Email: ali@gmail.com
NIC: 3520212345671
Age: 24

Name: faheem
Phone Number: 03001234567
Email: fah@gmail.com
NIC: 3520212345671
Age: 100

```
ADMIN MENU

1. View Driver Records
2. View Riders Records
3. View Feedback
4. Main Menu / Exit

Choose an option: 2

REGISTERED RIDERS

Name: faheem
Phone Number: 03121234567
Email: asc@gmail.com
NIC: 1212121212121
Age: 20

Name: ali
Phone Number: 03001234567
Email: ali@gmail.com
NIC: 1234567890123
Age: 24

Name: faheem
Phone Number: 03001234567
Email: fa@gmail.com
NIC: 3520212345671
Age: 18

Name: faheen
Phone Number: 12345678910
Email: faheem@gmail.com
NIC: 3520212345671
Age: 19

Name: faheem
Phone Number: 12345678910
Email: fah@gmail.com
NIC: 3520212345678
Age: 19
```

6. Random Testing

Test Case ID	Module	Random Input	Expected Output	Actual Output	Status
RT-01	Main Menu	Option = 7	Invalid option message	“Incorrect option” displayed	Pass
RT-02	Admin Login	Username=random123, Password=abc	Login denied	“Wrong info” displayed	Pass
RT-03	Driver Registration	Name = 123Ali	Invalid name warning	Validation message displayed	Pass
RT-04	Driver Registration	Phone = 98765	Reject invalid phone length	Error displayed	Pass
RT-05	Driver Registration	Email = abcgmail.com	Reject invalid email	Error displayed	Pass

RT-06	Driver Registration	Age = 15	Registration denied	“Not old enough” shown	Pass
RT-07	Driver Registration	NIC = 12345	Reject invalid NIC	Error displayed	Pass
RT-08	Rider Registration	Name = @Ahmed	Reject invalid name	Validation worked	Pass
RT-09	Rider Registration	Phone = random text	Reject invalid phone	Error displayed	Pass
RT-10	Rider Registration	Email = test@	Reject invalid email	Error displayed	Pass
RT-11	Ride Booking	Passengers = 20	No suitable car available	“No car available” shown	Pass
RT-12	Ride Booking	Pickup = 20	Invalid location warning	“Incorrect location” shown	Pass
RT-13	Ride Booking	Destination = -2	Invalid destination warning	Error displayed	Pass
RT-14	Ride Booking	Pickup=1, Destination=10	Fare calculation	Correct fare displayed	Pass
RT-15	Ride Cancellation	Random Ride ID = 999999	Ride not found	“Ride ID not found” shown	Pass
RT-16	Feedback	Rating = 9	Reject invalid rating	Error displayed	Module not working
RT-17	Feedback	Rating = 5	Accept feedback	Feedback stored successfully	Module not working
RT-18	File Handling	Missing file name	File error message	“File doesnt exist” shown	Pass
RT-19	Rider Menu	Option = 99	Invalid option message	Correct warning displayed	Pass
RT-20	Driver Assignment	No drivers file	Default driver assigned	“Huma” returned	Pass

Welcome to Elite Ride!
Thank you for choosing us

MAIN MENU

1. Rider
2. Driver
3. Admin

Choose an option: 7

Incorrect option.

Welcome to Elite Ride!
Thank you for choosing us

MAIN MENU

1. Rider
2. Driver
3. Admin

Choose an option: █

MAIN MENU

1. Rider
2. Driver
3. Admin

Choose an option: 3

LOG IN

Enter username: random123

Enter password: abc

Wrong info, please try again!

LOG IN

Enter username:

Welcome to Elite Ride!
Thank you for choosing us

MAIN MENU

1. Rider
2. Driver
3. Admin

Choose an option: 2

Register yourself as a driver!

Enter your name: 123ali
invalid name! name must only be alphabetic
Enter Name:

Choose an option: 2

Register yourself as a driver!

Enter your name: 123ali
invalid name! name must only be alphabetic
Enter Name: daud
Enter Phone Number: 98765
Wrong number! Length must be 11 digits
Enter Phone Numbe:

You are Registered!

Lets book a ride for you!

Enter the number of passengers: 20
No car available for given number of passengers!

RIDER MENU

1. Search for another ride
2. See all accepted rides
3. Cancel a ride
4. Main Menu / Exit

Choose an option:

A White coloured Mercedes will be waiting right at your location!

- 1)University Road
- 2)Karsaz
- 3)Bahadurabad
- 4)Gulshan
- 5)North Nazimabad
- 6)Baloch Colony
- 7)Defense
- 8)Clifton
- 9)Metropole
- 10)Kemari
- 11)Saddar
- 12)Gulberg
- 13)Lyari Expressway

Enter a number for Pickup location(1-13): 20

Incorrect location

Enter a number for Pickup location(1-13): 2

Enter a number for Destination(1-13): 20

Incorrect location

Enter a number for Destination(1-13): _

A White coloured Mercedes will be waiting right at your location!

- 1)University Road
- 2)Karsaz
- 3)Bahadurabad
- 4)Gulshan
- 5)North Nazimabad
- 6)Baloch Colony
- 7)Defense
- 8)Clifton
- 9)Metropole
- 10)Kemari
- 11)Saddar
- 12)Gulberg
- 13)Lyari Expressway

Enter a number for Pickup location(1-13): 20

Incorrect location

Enter a number for Pickup location(1-13): 2

Enter a number for Destination(1-13): 20

Incorrect location

Enter a number for Destination(1-13): 10

Your car is booked!

Your Drivers name: faheem

Ride ID: 4030

The total fare from Karsaz to Kemari is 1300Rs

```

RIDER MENU

1. Search for another ride
2. See all accepted rides
3. Cancel a ride
4. Main Menu / Exit

Choose an option: 999999

Incorrect option! Please try again.

RIDER MENU

1. Search for another ride
2. See all accepted rides
3. Cancel a ride
4. Main Menu / Exit

Choose an option: _
```

White Box Testing

1. Statement Coverage

1. Admin Module Coverage

Functions:

- login()
- view_feedback()
- view_driver_records()
- view_rider_records()
- admin_menu()

Required Test Cases

TC1: Valid Admin Login

Input:

- username = dsaa
- password = dsaa.project

Covers:

- if (correct credentials) branch
- success message statement

```

      Welcome to Elite Ride!
      Thank you for choosing us

      MAIN MENU

      1. Rider
      2. Driver
      3. Admin

      Choose an option: 3

      LOG IN
Enter username: dsaa
Enter password: dsaa.project
You're logged in!

      ADMIN MENU

      1. View Driver Records
      2. View Riders Records
      3. View Feedback
      4. Main Menu / Exit

      Choose an option:
```

TC2: Invalid Login (forces recursion)

Input:

- wrong username/password

Covers:

- if (username!=user || password!=pw)
- recursive login() call

```

Welcome to Elite Ride!
Thank you for choosing us

MAIN MENU

1. Rider
2. Driver
3. Admin

Choose an option: 3

LOG IN
Enter username: shawal
Enter password: faheem
Wrong info, please try again!

LOG IN
Enter username: _
```

TC3: Admin Menu Options

Run each option:

1. View driver records
2. View rider records
3. View feedback
4. Exit

Covers:

- all switch cases (1–4)
- default case

ADMIN MENU

1. View Driver Records
2. View Riders Records
3. View Feedback
4. Main Menu / Exit

Choose an option: 1

DRIVER RECORDS

Name: faheem
Phone Number: 03121234567
Email: asc@gmail.com
NIC: 1212121212121
Age: 20

Name: faheem
Phone Number: 03@12345678
Email: f@gmail.com
NIC: 3520212345671
Age: 32

Name: faheem
Phone Number: 03123456789
Email: ali@gmail.com
NIC: abcdef1234567
Age: 24

Name: faheem
Phone Number: 03123456789
Email: ali@gmail.com
NIC: 3520212345671
Age: 24

Name: faheem
Phone Number: 03001234567
Email: fah@gmail.com
NIC: 3520212345671
Age: 100

ADMIN MENU

1. View Driver Records
2. View Riders Records
3. View Feedback
4. Main Menu / Exit

Choose an option: 2

REGISTERED RIDERS

Name: faheem
Phone Number: 03121234567
Email: asc@gmail.com
NIC: 1212121212121
Age: 20

Name: ali
Phone Number: 03001234567
Email: ali@gmail.com
NIC: 1234567890123
Age: 24

Name: faheem
Phone Number: 03001234567
Email: fa@gmail.com
NIC: 3520212345671
Age: 18

Name: faheen
Phone Number: 12345678910
Email: faheem@gmail.com
NIC: 3520212345671
Age: 19

Name: faheem
Phone Number: 12345678910
Email: fah@gmail.com
NIC: 3520212345678
Age: 19

```
Name: ayb
Phone Number: 12121212121
Email: d@.com
NIC: 3520212345678
Age: 32
```

Press any key to continue

ADMIN MENU

1. View Driver Records
2. View Riders Records
3. View Feedback
4. Main Menu / Exit

Choose an option: 3

REVIEWS

Press any key to continue

ADMIN MENU

1. View Driver Records
2. View Riders Records
3. View Feedback
4. Main Menu / Exit

Choose an option:

2. Driver Module Coverage

Function: driver_register()

TC4: Valid Driver Registration

Input:

- name = "Ali"
- phone = "03001234567"
- email = "ali@gmail.com"
- age = 25
- NIC = "1234567890123"

Covers:

- isName(name) == true
- phone length valid

- email validation success
- $\text{age} \geq 18$ branch
- NIC validation
- file writing statements
- return true

```
MAIN MENU

1. Rider
2. Driver
3. Admin

Choose an option: 2
Register yourself as a driver!
Enter your name: Ali
Enter Phone Number: 03001234567
Enter Email: ali@gmail.com
Enter Age: 25
Enter NIC: 1234567890123
You have been registered!

1. Main menu
0. Exit

Choose an option: _
```

TC5: Invalid Name

Input:

- name = "Ali123"

Covers:

- while (!isName(name))

```
MAIN MENU

1. Rider
2. Driver
3. Admin

Choose an option: 2

Register yourself as a driver!

Enter your name: Ali123
invalid name! name must only be alphabetic
Enter Name:
```

TC6: Invalid Phone Number

Input:

- phone = "12345"

Covers:

- phone length loop

```
MAIN MENU

1. Rider
2. Driver
3. Admin

Choose an option: 2

Register yourself as a driver!

Enter your name: Ali123
invalid name! name must only be alphabetic
Enter Name: Ali
Enter Phone Number: 12345
Wrong number! Length must be 11 digits
Enter Phone Numbe:
```

TC7: Underage Driver

Input:

- age = 16

Covers:

- else { Not old enough }

```

MAIN MENU

1. Rider
2. Driver
3. Admin

Choose an option: 2

Register yourself as a driver!

Enter your name: Ali123
invalid name! name must only be alphabetic
Enter Name: Ali
Enter Phone Number: 12345
Wrong number! Length must be 11 digits
Enter Phone Numbe: 12345678910
Enter Email: fah@gmail.com
Enter Age: 16
Not old enough to register

1. Main menu
0. Exit

Choose an option: _

```

3. Rider Module Coverage

Function: Register(), get_ride()

TC8: Valid Rider Registration + Ride Booking

Input:

- valid name, phone, email, age ≥ 18

Covers:

- registration success block
- file writing to Registeredriders.txt
- calls get_ride()

```
        Choose an option: 1
Please register yourself first
Enter Name: ali
Enter Phone Number: 03001234567
Enter Email: acc@gmail.com
Enter Age: 21
Enter NIC: 1234567890123

You are Registered!

Lets book a ride for you!

Enter the number of passengers: 4

A Silver coloured Hyundai will be waiting right at your location!

1)University Road
2)Karsaz
3)Bahadurabad
4)Gulshan
5)North Nazimabad
6)Baloch Colony
7)Defense
8)Clifton
9)Metropole
10)Kemari
11)Saddar
12)Gulberg
13)Lyari Expressway

Enter a number for Pickup location(1-13): 2
Enter a number for Destination(1-13): 8

        Your car is booked!

Your Drivers name: faheem
Ride ID: 12890
The total fare from Karsaz to Clifton is 1400Rs
```

TC9: Invalid Email

Input:

- email = "abc"

Covers:

- while(!is_valid(email))

```
        Welcome to Elite Ride!  
        Thank you for choosing us
```

```
        MAIN MENU
```

- ```
 1. Rider
 2. Driver
 3. Admin
```

```
 Choose an option: 1
```

```
Please register yourself first
```

```
Enter Name: ali
```

```
Enter Phone Number: 03001234567
```

```
Enter Email: abc
```

```
Wrong email provided!
```

```
Enter correct Email:
```

### TC10: Underage Rider

#### Input:

- age = 15

#### Covers:

- else { Not old enough }

```
 Welcome to Elite Ride!
 Thank you for choosing us
```

```
 MAIN MENU
```

- ```
        1. Rider
        2. Driver
        3. Admin
```

```
        Choose an option: 1
```

```
Please register yourself first
```

```
Enter Name: ali
```

```
Enter Phone Number: 03001234567
```

```
Enter Email: abc
```

```
Wrong email provided!
```

```
Enter correct Email: acc@gmail.com
```

```
Enter Age: 15
```

```
Not old enough to register!
```

- ```
 1. Main menu
 0. Exit
```

```
 Choose an option:
```

#### 4. Car Assignment Module

**Function:** assignCar(int n)

**TC11:** Valid Passenger Match

**Input:**

- passengers = 4

**Covers:**

- if (capacity\_of\_passengers >= n)
- success return true



```
Enter the number of passengers: 4

A Silver coloured Hyundai will be waiting right at your location!

1)University Road
2)Karsaz
3)Bahadurabad
4)Gulshan
5)North Nazimabad
6)Baloch Colony
7)Defense
8)Clifton
9)Metropole
10)Kemari
11)Saddar
12)Gulberg
13)Lyari Expressway

Enter a number for Pickup location(1-13): 2
Enter a number for Destination(1-13): 8

 Your car is booked!

Your Drivers name: faheem
Ride ID: 12890
The total fare from Karsaz to Clifton is 1400Rs
```

### TC12: No Car Available

#### Input:

- passengers = 100

#### Covers:

- loop ends without match
- "No car available" statement

```

RIDER MENU

1. Search for another ride
2. See all accepted rides
3. Cancel a ride
4. Main Menu / Exit

Choose an option: 1

Lets book a ride for you!

Enter the number of passengers: 100
No car available for given number of passengers!

Press any key to continue

```

## 5. Graph + Billing (Dijkstra)

### TC14: Valid Route Fare Calculation

#### Input:

- pickup = 1
- destination = 8

#### Covers:

- graph construction
- edge insertion
- Dijkstra loop
- distance update
- output statement

```
Enter the number of passengers: 6

A Black coloured BMW will be waiting right at your location!

1)University Road
2)Karsaz
3)Bahadurabad
4)Gulshan
5)North Nazimabad
6)Baloch Colony
7)Defense
8)Clifton
9)Metropole
10)Kemari
11)Saddar
12)Gulberg
13)Lyari Expressway

Enter a number for Pickup location(1-13): 2
Enter a number for Destination(1-13): 4

 Your car is booked!

Your Drivers name: faheem
Ride ID: 13419
The total fare from Karsaz to Gulshan is 600Rs
```

## 6. File Handling Coverage

### TC15: File Read/Write

#### Triggers:

- driver registration
- rider registration
- feedback
- admin viewing files

#### Covers:

- fstream open
- ios::app
- getline
- file not found condition

```
1. View Driver Records
2. View Riders Records
3. View Feedback
4. Main Menu / Exit

Choose an option: 2

REGISTERED RIDERS

Name: faheem
Phone Number: 03121234567
Email: asc@gmail.com
NIC: 1212121212121
Age: 20

Name: ali
Phone Number: 03001234567
Email: ali@gmail.com
NIC: 1234567890123
Age: 24

Name: faheem
Phone Number: 03001234567
Email: fa@gmail.com
NIC: 3520212345671
Age: 18

Name: faheen
Phone Number: 12345678910
Email: faheem@gmail.com
NIC: 3520212345671
Age: 19

Name: faheem
Phone Number: 12345678910
Email: fah@gmail.com
NIC: 3520212345678
Age: 19
```

## 2. Branch Coverage

### 1. Admin Module

**Function:** login()

**Branch 1:** Credential Check

**Condition:** if (username != user || password != pw)

| Test Case ID | Input (Username, Password) | Condition Result | Expected Output  | Branch      |
|--------------|----------------------------|------------------|------------------|-------------|
| TC01         | dsaa, dsaa.project         | FALSE            | Login successful | ELSE branch |

|      |              |      |                          |           |
|------|--------------|------|--------------------------|-----------|
| TC02 | wrong, wrong | TRUE | Wrong info + retry login | IF branch |
|------|--------------|------|--------------------------|-----------|

```

MAIN MENU

1. Rider
2. Driver
3. Admin

Choose an option: 3

LOG IN
Enter username: dsaa
Enter password: dsa123
Wrong info, please try again!

LOG IN
Enter username: dsaa
Enter password: dsaa.project
You're logged in!

ADMIN MENU

1. View Driver Records
2. View Riders Records
3. View Feedback
4. Main Menu / Exit

Choose an option:

```

## 2. Admin Menu

**Function:** admin\_menu()

**Condition:** switch(option)

| Test Case ID | Input | Expected Output     | Branch         |
|--------------|-------|---------------------|----------------|
| TC03         | 1     | View Driver Records | Case 1         |
| TC04         | 2     | View Rider Records  | Case 2         |
| TC05         | 3     | View Feedback       | Case 3         |
| TC06         | 4     | Exit                | Case 4         |
| TC07         | 9     | Invalid option      | Default branch |



# ADMIN MENU

1. View Driver Records
2. View Riders Records
3. View Feedback
4. Main Menu / Exit

Choose an option: 1

## DRIVER RECORDS

Name: faheem  
Phone Number: 03121234567  
Email: asc@gmail.com  
NIC: 1212121212121  
Age: 20

Name: faheem  
Phone Number: 03@12345678  
Email: f@gmail.com  
NIC: 3520212345671  
Age: 32

Name: faheem  
Phone Number: 03123456789  
Email: ali@gmail.com  
NIC: abcdef1234567  
Age: 24

Name: faheem  
Phone Number: 03123456789  
Email: ali@gmail.com  
NIC: 3520212345671  
Age: 24

Name: faheem  
Phone Number: 03001234567  
Email: fah@gmail.com  
NIC: 3520212345671  
Age: 100

# ADMIN MENU

1. View Driver Records
2. View Riders Records
3. View Feedback
4. Main Menu / Exit

Choose an option: 2

## REGISTERED RIDERS

Name: faheem  
Phone Number: 03121234567  
Email: asc@gmail.com  
NIC: 1212121212121  
Age: 20

Name: ali  
Phone Number: 03001234567  
Email: ali@gmail.com  
NIC: 1234567890123  
Age: 24

Name: faheem  
Phone Number: 03001234567  
Email: fa@gmail.com  
NIC: 3520212345671  
Age: 18

Name: faheen  
Phone Number: 12345678910  
Email: faheem@gmail.com  
NIC: 3520212345671  
Age: 19

Name: faheem  
Phone Number: 12345678910  
Email: fah@gmail.com  
NIC: 3520212345678  
Age: 19



Name: ayb  
Phone Number: 12121212121  
Email: d@.com  
NIC: 3520212345678  
Age: 32

Press any key to continue

#### ADMIN MENU

1. View Driver Records
2. View Riders Records
3. View Feedback
4. Main Menu / Exit

Choose an option: 3

#### REVIEWS

Press any key to continue

#### ADMIN MENU

1. View Driver Records
2. View Riders Records
3. View Feedback
4. Main Menu / Exit

Choose an option:

```
ADMIN MENU

1. View Driver Records
2. View Riders Records
3. View Feedback
4. Main Menu / Exit

Choose an option: 9

Incorrect option! Please try again.

ADMIN MENU

1. View Driver Records
2. View Riders Records
3. View Feedback
4. Main Menu / Exit

Choose an option: 4

1. Main menu
0. Exit

Choose an option:
```

### 3. Driver Registration

**Function:** driver\_register()

**Branch 1: Name Validation**

**Condition:** while (!isName(name))

| TC ID | Input  | Condition Result | Expected Output | Branch    |
|-------|--------|------------------|-----------------|-----------|
| TC08  | Ali123 | TRUE             | Re-prompt input | Loop      |
| TC09  | Ali    | FALSE            | Accept name     | Exit loop |

**Branch 2: Phone Validation**

**Condition:** while (phone\_number.length() != 11)

| TC ID | Input       | Result | Branch |
|-------|-------------|--------|--------|
| TC10  | 12345       | TRUE   | Loop   |
| TC11  | 03001234567 | FALSE  | Exit   |

### Branch 3: Email Validation

**Condition:** while(!is\_valid(email))

| TC ID | Input         | Result | Branch |
|-------|---------------|--------|--------|
| TC12  | abcm.com      | TRUE   | Loop   |
| TC13  | abc@gmail.com | FALSE  | Exit   |

### Branch 4: Age Condition

**Condition:** if (age >= 18)

| TC ID | Age | Result | Branch          |
|-------|-----|--------|-----------------|
| TC14  | 20  | TRUE   | Register driver |
| TC15  | 16  | FALSE  | Not eligible    |

### Branch 5: NIC Validation

**Condition:** while (NIC.length() != 13)

| TC ID | Input         | Result | Branch |
|-------|---------------|--------|--------|
| TC16  | 12345         | TRUE   | Loop   |
| TC17  | 1234567890123 | FALSE  | Exit   |

## 4. Rider Registration

**Function:** Register()

### Branch 1: Age Check

**Condition:** if (age >= 18)

| TC ID | Age | Expected Output | Branch |
|-------|-----|-----------------|--------|
| TC18  | 22  | Registered      | TRUE   |
| TC19  | 16  | Not allowed     | FALSE  |

### Branch 2: Phone Validation

**Condition:** while(length != 11)

| TC ID | Input   | Branch |
|-------|---------|--------|
| TC20  | invalid | TRUE   |
| TC21  | valid   | FALSE  |

### Branch 3: Email Validation

**Condition:** while(!is\_valid(email))

| TC ID | Input         | Branch |
|-------|---------------|--------|
| TC22  | wrong email   | TRUE   |
| TC23  | correct email | FALSE  |

## 5. Ride Booking

**Function:** get\_ride()

### Branch 1: Car Assignment

**Condition:** if(assign == true)

| TC ID | Input (Passengers)               | Expected Output | Branch |
|-------|----------------------------------|-----------------|--------|
| TC24  | 2 passengers                     | Car assigned    | TRUE   |
| TC25  | 10 passengers (invalid capacity) | No car          | FALSE  |

### Branch 2: Pickup Validation

**Condition:** if(pickup > 13 || pickup < 1)

| TC ID | Input | Branch |
|-------|-------|--------|
| TC26  | 5     | FALSE  |
| TC27  | 20    | TRUE   |

### Branch 3: Destination Validation

**Condition:**  $\text{if}(d > 13 \parallel d < 1)$

| TC ID | Input | Branch |
|-------|-------|--------|
| TC28  | 3     | FALSE  |
| TC29  | 99    | TRUE   |

### 6. Car Assignment

**Function:** `assignCar(int n)`

**Condition:**  $\text{if}(\text{capacity\_of\_passengers} \geq n)$

| TC ID | Input         | Expected Output  | Branch |
|-------|---------------|------------------|--------|
| TC30  | 2 passengers  | Car assigned     | TRUE   |
| TC31  | 20 passengers | No car available | FALSE  |

### 7. File Handling

**Condition:**  $\text{if}(!\text{File})$

| TC ID | File Status  | Expected Output | Branch |
|-------|--------------|-----------------|--------|
| TC36  | File exists  | Display data    | FALSE  |
| TC37  | File missing | Error message   | TRUE   |

## 3.Condition Coverage

### 1. Admin Login Condition Coverage

**Condition:**

$\text{if}(\text{username} \neq \text{user} \parallel \text{password} \neq \text{pw})$

| TC ID | Username | Password     | username!=user | password!=pw | Expected Result | Condition Covered |
|-------|----------|--------------|----------------|--------------|-----------------|-------------------|
| TC01  | dsaa     | dsaa.project | FALSE          | FALSE        | Login Success   | Both FALSE        |
| TC02  | wrong    | dsaa.project | TRUE           | FALSE        | Login Failed    | First TRUE        |
| TC03  | dsaa     | wrong        | FALSE          | TRUE         | Login Failed    | Second TRUE       |
| TC04  | wrong    | wrong        | TRUE           | TRUE         | Login Failed    | Both TRUE         |

## 2. File Handling Condition Coverage

**Condition:**

if (!File)

| TC ID | File Status  | Condition Result | Expected Result | Condition Covered |
|-------|--------------|------------------|-----------------|-------------------|
| TC05  | File Exists  | FALSE            | File Opened     | FALSE             |
| TC06  | File Missing | TRUE             | Error Message   | TRUE              |

## 3. Character & Name Validation Coverage

**Condition:**

(c >= 'a' && c <= 'z') || (c >= 'A' && c <= 'Z')

| TC ID | Input | Lowercase | Uppercase | Expected Result | Condition Covered |
|-------|-------|-----------|-----------|-----------------|-------------------|
| TC07  | a     | TRUE      | FALSE     | Valid           | TRUE/FALSE        |
| TC08  | Z     | FALSE     | TRUE      | Valid           | FALSE/TRUE        |
| TC09  | 5     | FALSE     | FALSE     | Invalid         | FALSE/FALSE       |

## 4. Name Validation

**Condition:**

if (!isChar(name[i]))

| TC ID | Name Input | Condition Result | Expected Result | Condition Covered |
|-------|------------|------------------|-----------------|-------------------|
|-------|------------|------------------|-----------------|-------------------|

|      |        |       |              |       |
|------|--------|-------|--------------|-------|
| TC10 | Ali    | FALSE | Valid Name   | FALSE |
| TC11 | Ali123 | TRUE  | Invalid Name | TRUE  |

## 5. Email Validation Conditions

### Condition 1:

if (!isChar(email[0]))

| TC ID | Email                                              | Condition Result | Expected Result | Condition Covered |
|-------|----------------------------------------------------|------------------|-----------------|-------------------|
| TC12  | <a href="mailto:ali@gmail.com">ali@gmail.com</a>   | FALSE            | Valid           | FALSE             |
| TC13  | <a href="mailto:1ali@gmail.com">1ali@gmail.com</a> | TRUE             | Invalid         | TRUE              |

### Condition 2:

if (At == -1 || Dot == -1)

| TC ID | Email                                            | At==<br>-1 | Dot==<br>-1 | Expected Result | Condition Covered |
|-------|--------------------------------------------------|------------|-------------|-----------------|-------------------|
| TC14  | aligmail.com                                     | TRUE       | FALSE       | Invalid         | TRUE/FALSE        |
| TC15  | ali@gmailcom                                     | FALSE      | TRUE        | Invalid         | FALSE/TRUE        |
| TC16  | <a href="mailto:ali@gmail.com">ali@gmail.com</a> | FALSE      | FALSE       | Valid           | FALSE/FALSE       |

### Condition 3:

if (At > Dot)

| TC ID | Email                                            | Condition Result | Expected Result |
|-------|--------------------------------------------------|------------------|-----------------|
| TC17  | ali.gmail@com                                    | TRUE             | Invalid         |
| TC18  | <a href="mailto:ali@gmail.com">ali@gmail.com</a> | FALSE            | Valid           |

### Condition 4:

return !(Dot >= email.length()-1)

| TC ID | Email                                            | Condition Result | Expected Result |
|-------|--------------------------------------------------|------------------|-----------------|
| TC19  | ali@gmail.                                       | TRUE             | Invalid         |
| TC20  | <a href="mailto:ali@gmail.com">ali@gmail.com</a> | FALSE            | Valid           |

## 6. Driver Registration Coverage

### Condition:

if (age >= 18)

| TC ID | Age | Condition Result | Expected Result |
|-------|-----|------------------|-----------------|
| TC21  | 25  | TRUE             | Registered      |
| TC22  | 16  | FALSE            | Rejected        |

### Phone Validation:

while (phone\_number.length() != 11)

| TC ID | Phone       | Condition Result |
|-------|-------------|------------------|
| TC23  | 12345       | TRUE             |
| TC24  | 03123456789 | FALSE            |

### NIC Validation:

while (NIC.length() != 13)

| TC ID | NIC           | Condition Result |
|-------|---------------|------------------|
| TC25  | 12345         | TRUE             |
| TC26  | 1234567890123 | FALSE            |

## 7. Car Assignment Condition Coverage

### Condition:

if (capacity\_of\_passengers >= n)



| TC ID | Passengers | Condition Result | Expected Result |
|-------|------------|------------------|-----------------|
| TC27  | 4          | TRUE             | Car Assigned    |
| TC28  | 100        | FALSE            | No Car          |

## 8. Ride Booking Condition Coverage

### Condition:

if (assign)

| TC ID | Input               | Condition Result | Expected Result |
|-------|---------------------|------------------|-----------------|
| TC29  | Valid passengers    | TRUE             | Ride Booked     |
| TC30  | Too many passengers | FALSE            | Booking Failed  |

## 9. Location Validation Coverage

### Pickup Condition:

if (pickup > 13 || pickup < 1)

| TC ID | Input | pickup>13 | pickup<1 | Expected |
|-------|-------|-----------|----------|----------|
| TC31  | 20    | TRUE      | FALSE    | Invalid  |
| TC32  | 5     | FALSE     | FALSE    | Valid    |

### Destination Condition:

if (d > 13 || d < 1)

| TC ID | Input | Condition Result |
|-------|-------|------------------|
| TC33  | 15    | TRUE             |
| TC34  | 7     | FALSE            |

## 10. Feedback Condition Coverage

### Condition:

if (rating < 1 || rating > 5)

| TC ID | Rating | Condition Result | Expected |
|-------|--------|------------------|----------|
| TC35  | 0      | TRUE             | Invalid  |
| TC36  | 6      | TRUE             | Invalid  |
| TC37  | 4      | FALSE            | Valid    |

## 4. Loop Testing

### 1. Name Validation Loop

**Function:** isName()

for(int i=0;i<s;i++)

**Loop Type:** For Loop (Input Validation)

| Test Case ID | Input Name                  | Loop Iterations | Expected Result               | Case Type              |
|--------------|-----------------------------|-----------------|-------------------------------|------------------------|
| TC01         | "" (empty)                  | 0               | Accepted as valid (edge case) | Zero iteration         |
| TC02         | "A"                         | 1               | Valid                         | Single iteration       |
| TC03         | "Ali"                       | 3               | Valid                         | Normal                 |
| TC04         | "Ali123"                    | 6               | Invalid                       | Mixed input            |
| TC05         | "ABCDEFGHJKLMNOPQRSTUVWXYZ" | 26              | Valid                         | Maximum iteration      |
| TC06         | "A!i"                       | 3               | Invalid                       | Special char detection |

### 2. Phone Number Validation

**Function:**

while(phone\_number.length() != 11)

**Loop Type:** While Loop

| Test Case ID | Input                    | Loop Iterations | Expected Result          | Case Type      |
|--------------|--------------------------|-----------------|--------------------------|----------------|
| TC07         | "03012345678"            | 0               | Accepted                 | Valid input    |
| TC08         | "123" →<br>"03012345678" | 1+              | Correct after retry      | Single failure |
| TC09         | "" → valid               | multiple        | Correct after correction | Boundary loop  |
| TC10         | "11111111111"            | 0               | Accepted                 | Exact boundary |
| TC11         | "123456789"              | multiple        | Reject until fixed       | Invalid length |

### 3. Email Validation Loop

#### Function:

```
while(!is_valid(email))
```

#### Loop Type: While Loop

| Test Case ID | Input Email                                      | Loop Iterations | Expected Result      | Case Type      |
|--------------|--------------------------------------------------|-----------------|----------------------|----------------|
| TC12         | " <a href="mailto:a@gmail.com">a@gmail.com</a> " | 0               | Valid                | Correct input  |
| TC13         | "abc" → valid                                    | 1+              | Accepted after retry | Invalid first  |
| TC14         | "@gmail.com"                                     | multiple        | Rejected             | Missing name   |
| TC15         | "abcgmail.com"                                   | multiple        | Rejected             | Missing @      |
| TC16         | " <a href="mailto:abc@.com">abc@.com</a> "       | multiple        | Rejected             | Invalid format |

### 4. NIC Validation Loop

#### Function:

```
while(NIC.length() != 13)
```

#### Loop Type: While Loop

| Test Case ID | Input NIC       | Loop Iterations | Expected Result      | Case Type     |
|--------------|-----------------|-----------------|----------------------|---------------|
| TC17         | "1234567890123" | 0               | Valid                | Correct input |
| TC18         | "12345" → valid | 1+              | Accepted after retry | Invalid first |
| TC19         | "" → valid      | multiple        | Fixed input accepted | Empty input   |

|      |                  |          |                      |                |
|------|------------------|----------|----------------------|----------------|
| TC20 | "11111111111111" | 0        | Valid                | Boundary       |
| TC21 | "123456789"      | multiple | Reject until correct | Invalid length |

## 5. Driver Name File Reading Loop

### Function:

```
while(!File.eof())
```

### Loop Type: File Reading Loop

| Test Case ID | File State             | Loop Iterations | Expected Result | Case Type        |
|--------------|------------------------|-----------------|-----------------|------------------|
| TC22         | Empty file             | 0               | No output       | Zero iteration   |
| TC23         | 1 line file            | 1               | Read line       | Single iteration |
| TC24         | 5 line file            | 5               | Read all lines  | Normal           |
| TC25         | Large file (100 lines) | 100             | Full read       | Stress test      |

## 6. Driver Registration Validation Loops

### Function:

```
while (!isName(name))
```

```
while (phone_number.length() != 11)
```

```
while (!is_valid(email))
```

```
while (NIC.length() != 13)
```

### Combined Loop Testing

| Test Case ID | Input Type         | Loop Behavior             | Expected Result         |
|--------------|--------------------|---------------------------|-------------------------|
| TC26         | All valid inputs   | 0 iterations in all loops | Successful registration |
| TC27         | Invalid name only  | name loop repeats         | Fix required            |
| TC28         | Invalid phone only | phone loop repeats        | Fix required            |
| TC29         | Invalid email only | email loop repeats        | Fix required            |
| TC30         | Invalid NIC only   | NIC loop repeats          | Fix required            |
| TC31         | All invalid        | multiple loop executions  | multiple corrections    |

7. Dijkstra Algorithm Loop

Function:

```
while (!pq.empty())
```

Loop Type: Priority Queue Loop

| Test Case ID | Graph Size   | Loop Iterations     | Expected Result  |
|--------------|--------------|---------------------|------------------|
| TC36         | Single node  | 1                   | Direct return    |
| TC37         | Small graph  | few iterations      | shortest path    |
| TC38         | Medium graph | moderate iterations | correct result   |
| TC39         | Dense graph  | many iterations     | optimized path   |
| TC40         | Large graph  | max iterations      | performance test |

5. CONTROL FLOW TESTING

Diagram: [CFG Diagram.jpg](#)

1. Control Flow Graph (CFG) Components

1.1 Nodes Identification

| Node ID | Description                    |
|---------|--------------------------------|
| N1      | main()                         |
| N2      | App::main_menu()               |
| N3      | Rider module                   |
| N4      | Driver module                  |
| N5      | Admin login                    |
| N6      | Admin menu                     |
| N7      | File viewing function          |
| N8      | Driver registration validation |
| N9      | Rider registration validation  |
| N10     | Ride booking (get_ride)        |
| N11     | Car assignment                 |
| N12     | Bill calculation (Dijkstra)    |

|     |                        |
|-----|------------------------|
| N13 | Linked list operations |
| N14 | Feedback system        |
| N15 | Exit                   |

## 1.2 Control Flow Edges

| From Node       | To Node                               | Condition / Decision        |
|-----------------|---------------------------------------|-----------------------------|
| N1 → N2         | main → menu                           | Program start               |
| N2 → N3         | if (option == 1)                      | Rider selected              |
| N2 → N4         | if (option == 2)                      | Driver selected             |
| N2 → N5         | if (option == 3)                      | Admin selected              |
| N5 → N6         | if (username & password correct)      | Login success               |
| N5 → N5         | if (username != password)             | Recursive retry login       |
| N3 → N9         | if (age >= 18)                        | Rider registration success  |
| N3 → N15        | if (age < 18)                         | Registration failed         |
| N4 → N8         | if (age >= 18)                        | Driver registration success |
| N4 → N15        | if (age < 18)                         | Driver rejected             |
| N9 → N10        | if (registration == true)             | Proceed to booking          |
| N10 → N11       | if (car found capacity >= passengers) | Assign car                  |
| N10 → N10       | if (car not found)                    | Retry booking loop          |
| N11 → N12       | if (ride confirmed)                   | Calculate fare (Dijkstra)   |
| N6 → N7         | switch-case option 1–3                | View records                |
| N6 → N6         | default case                          | Invalid option retry        |
| N7 → N13        | while(file not EOF)                   | Read file content           |
| N13 → N13       | loop condition (!eof)                 | File reading loop           |
| N10 → N14       | feedback submitted                    | Store feedback              |
| N3/N4/N10 → N15 | exit option                           | Program termination         |

## 2. Independent Execution Paths

| Path ID | Execution Flow                                   |
|---------|--------------------------------------------------|
| P1      | N1 → N2 → N3 → N9 → N10 → N11 → N12 → N15        |
| P2      | N1 → N2 → N4 → N8 → N15                          |
| P3      | N1 → N2 → N5 → N6 → N7 → N6 → N15                |
| P4      | N1 → N2 → N3 → N9 → N10 → N11 → N10 (loop retry) |

|    |                                             |
|----|---------------------------------------------|
| P5 | N1 → N2 → N5 → N5 → N6 (login failure loop) |
| P6 | N1 → N2 → N3 → N14 → N10 → N15              |

### 3. Loop Testing Analysis

#### 3.1 Identified Loops

| Loop ID | Location               | Type                    |
|---------|------------------------|-------------------------|
| L1      | Admin login()          | Recursive loop          |
| L2      | admin_menu()           | Menu loop               |
| L3      | rider_menu()           | Menu loop               |
| L4      | file() function        | File reading loop       |
| L5      | Linked list traversal  | While loop              |
| L6      | Car assignment         | While loop              |
| L7      | Dijkstra algorithm     | Priority queue loop     |
| L8      | Input validation loops | Name, email, NIC checks |
| L9      | Ride retry loop        | get_ride recursion      |

#### 3.2 Loop Testing Cases

##### L1: Admin Login Loop

| Test Case ID | Input Condition         | Expected Output      |
|--------------|-------------------------|----------------------|
| TC-L1-1      | Wrong username/password | Login retry          |
| TC-L1-2      | Multiple wrong attempts | Continuous recursion |
| TC-L1-3      | Correct credentials     | Exit loop            |

##### L2: Admin Menu Loop

| Test Case ID | Input          | Expected Output |
|--------------|----------------|-----------------|
| TC-L2-1      | 1              | Driver records  |
| TC-L2-2      | 2              | Rider records   |
| TC-L2-3      | 3              | Feedback        |
| TC-L2-4      | Invalid option | Menu repeat     |

|         |   |      |
|---------|---|------|
| TC-L2-5 | 4 | Exit |
|---------|---|------|

### **L3: Rider Menu Loop**

| Test Case ID | Input         | Expected Output |
|--------------|---------------|-----------------|
| TC-L3-1      | Book ride     | Ride created    |
| TC-L3-2      | View rides    | List displayed  |
| TC-L3-3      | Cancel ride   | Node deleted    |
| TC-L3-4      | Invalid input | Retry menu      |
| TC-L3-5      | Exit          | Stop execution  |

### **L4: File Reading Loop**

| Test Case ID | File State   | Expected Output |
|--------------|--------------|-----------------|
| TC-L4-1      | File exists  | Display content |
| TC-L4-2      | File empty   | No output       |
| TC-L4-3      | File missing | Error message   |

### **L5: Car Assignment Loop**

| Test Case ID | Input           | Expected Output        |
|--------------|-----------------|------------------------|
| TC-L6-1      | 1–3 passengers  | Small vehicle assigned |
| TC-L6-2      | 4–10 passengers | Medium vehicle         |
| TC-L6-3      | High passengers | No car available       |

### **L6: Dijkstra Loop**



| Test Case ID | Scenario                | Expected Output       |
|--------------|-------------------------|-----------------------|
| TC-L7-1      | Valid route             | Correct shortest path |
| TC-L7-2      | Same source/destination | 0 cost                |
| TC-L7-3      | No path                 | INT_MAX result        |

### L7: Input Validation Loops

| Test Case ID | Input         | Expected Output |
|--------------|---------------|-----------------|
| TC-L8-1      | Invalid name  | Retry input     |
| TC-L8-2      | Invalid email | Retry input     |
| TC-L8-3      | Invalid NIC   | Retry input     |

## 6. Cyclomatic Complexity Analysis

### 1. Function-wise Cyclomatic Complexity

#### 1. file(string name)

Decision points:

- `if(!File) → 1`
- `while(!File.eof()) → 1`

$$CC = 2 + 1 = 3$$

#### 2. Admin::login()

Decision points:

- `if (username!=user || password!=pw) → 1`
- recursion call in if path → (usually not extra node, but adds path)
- else → 0 extra

$$CC = 2$$

#### 3. Admin::admin\_menu()

**Decision points:**

- `switch(option)` → 1
- `case 1,2,3,4` → 4 branches
- `default` → 1 extra branch
- recursion calls inside cases (not always counted)

$$CC = 1 + 5 = 6$$

**4. `isName(string name)`****Decision points:**

- `for` loop → 1
- `if (!isChar(...))` → 1

$$CC = 3$$

**5. `is_valid(string email)`****Decision points:**

- `if (!isChar(email[0]))` → 1
- `for` loop → 1
- `if (email[i]=='@')` → 1
- `else if (email[i]=='.)` → 1
- `if (At == -1 || Dot == -1)` → 1
- `if (At > Dot)` → 1
- final return condition → 1

$$CC = 7$$

**6. `Driver::driver_register()`****Decision points:**

- `while (!isName(name))` → 1
- `while(phone_number.length()!=11)` → 1
- `while(!is_valid(email))` → 1

- if (age >= 18) → 1
- while (NIC.length()!=13) → 1
- else → implicit branch

**CC = 6**

## **7. dijkstra(Graph g, int start)**

**Decision points:**

- while(!pq.empty()) → 1
- for (auto e : g.adj[u]) → 1
- if(dist[v] > dist[u] + weight) → 1

**CC = 4**

## **8. CarList::addCar()**

**Decision points:**

- if(head==NULL) → 1
- else → 1

**CC = 2**

## **9. CarList::assignCar(int n)**

**Decision points:**

- while(temp!=NULL) → 1
- if(temp->capacity\_of\_passengers >= n) → 1

**CC = 3**

## **10. RideLinkedList::Add\_ride()**

**Decision points:**

- if(head==NULL) → 1
- else → 1
- while(current->next!=NULL) → 1

**CC = 3**

### **11. RideLinkedList::Display\_ride()**

**Decision points:**

- `if(head==NULL) → 1`
- `while(current!=NULL) → 1`

**CC = 3**

### **12. RideLinkedList::delete\_ride()**

**Decision points:**

- `if(head!=NULL) → 1`
- `if(head->data==d) → 1`
- `else → 1`
- `while(current!=NULL && current->data!=d) → 1`
- `if(current!=NULL) → 1`
- `else → 1`
- `nested conditions → +1`

**CC = 7**

### **13. Rider::Register()**

**Decision points:**

- `while(!isName(name)) → 1`
- `while(phone_number.length()!=11) → 1`
- `while(!is_valid(email)) → 1`
- `if(age>=18) → 1`
- `while(NIC.length()!=13) → 1`
- `else → 1`

**CC = 6**

#### **14. Rider::give\_feedback()**

##### **Decision points:**

- $\text{if}(\text{rating} < 1 \parallel \text{rating} > 5) \rightarrow 1$
- recursion call  $\rightarrow 1$  logical path

**CC = 2**

#### **15. Rider::get\_driver()**

##### **Decision points:**

- $\text{if}(!\text{in}) \rightarrow 1$
- $\text{while}(\text{in} \gg \text{word}) \rightarrow 1$

**CC = 3**

#### **16. Rider::get\_ride()**

##### **Decision points:**

- $\text{if}(\text{assign}) \rightarrow 1$
- calls to pickup & destination validation  $\rightarrow$  (inside functions)
- indirect branching via functions

Main:

- $\text{if}(\text{assign}) \rightarrow 1$

**CC  $\approx$  2**

#### **17. Rider::pick\_up() / des\_loc()**

Each:

- $\text{if}(\text{pickup} > 13 \parallel \text{pickup} < 1) \rightarrow 1$
- recursion  $\rightarrow 1$

**CC = 2**

#### **18. Rider::rider\_menu()**

##### **Decision points:**

- switch(option) → 1
- 4 cases → 4
- default → 1

**CC = 6**

### **19. App::main\_menu()**

#### **Decision points:**

- switch(option) → 1
- 3 cases → 3
- default → 1

**CC = 5**

### **20. main()**

#### **Decision points:**

- if(choice=="1") → 1

**CC = 2**

## **2. Overall System Complexity**

If we sum major modules:

- Admin = 10–12
- Rider = 25–30
- Driver = 6
- Graph/Dijkstra = 4
- Linked lists = 10
- App/Main = 7

**Estimated Total Cyclomatic Complexity: 60–70**